Ahmet Soner Güleç

Aircraft Maintenance Engineer

Email : ahmetsonergulec@hotmail.com
Phone : +90 553 715 4169

Education

- **Eskişehir Technical University** — Eskişehir, Turkey
  B.S. Aircraft Maintenance Engineering / Avioncics; GPA: 3.1 — *Sep. 2020 – June 2026*

Experience

- **Artificial Intelligence in Aviation Research Laboratory** — Eskişehir, Turkey
  Project Specialist — *Oct. 2024 – Present*
  - **UAV System Design**: Developing flight control algorithms for unmanned aerial vehicles and avionics systems; managing sensor data processing and autonomous decision-making mechanisms.
  - **Simulation & Modeling**: Modeling aircraft system behaviors using MATLAB/Simulink and Python environments; conducting virtual testing and validation analysis of developed software.
  - **Hardware Development**: Designing electronic circuit boards to support embedded artificial intelligence systems.

- **Vilnius Gediminas Technical University** — Vilnius, Lithuania
  Aircraft Maintenance Engineering Intern — *Oct. 2023 – Apr. 2024*
  - **Avionics System Maintenance**: Execution of disassembly, assembly, and testing procedures for navigation, autopilot, and flight instrument systems on Airbus A320 and Cessna 172 aircraft.
  - **Design & Analysis**: Mechanical design of a quadcopter drone using SolidWorks and aerodynamic airfoil analysis of a fixed-wing UAV using XFLR5.
  - **Validation & Modeling**: Simulation and validation of system thermal behavior and performance using MATLAB and COMSOL.

Projects

- **FPV Drone**: Unmanned aerial vehicle capable of real-time target detection under electronic warfare and radar jamming conditions without relying on external navigation data.
- **TUSAŞ LIFT-UP Project**: Modeling and benchmarking performance datasets of a combat aircraft using machine learning techniques. (Also funded by **TÜBİTAK 2209-B**)
- **TÜBİTAK 2209-A**: Deep Learning-Based Bird Detection and Warning System designed to mitigate the operational impacts of bird strikes in aviation.
- **INFLOBOT**: AI-powered soft robotic system developed as an innovative solution for the aviation MRO sector.
- **BLDC Motor Driver Board**: Driver board featuring 2 kW / 48 V / 42 A, 97.15% efficiency, and FOC algorithm.
- **Isolation Monitoring System**: Real-time insulation resistance tracking system monitoring positive and negative lines via high-precision differential amplifiers.
- **Vehicle Control System**: Central system managing vehicle communication and control protocols.

Honors and Awards

- **2025 Teknofest Efficiency Challenge Electric Vehicle Races**: Hydromobile Category Finalist
- **2025 Shell Eco-marathon Europe and Africa**: Prototype Hydrogen Fuel Cell Category Finalist
- **2024 Teknofest Efficiency Challenge Electric Vehicle Races**: Hydromobile Category Finalist
- **2023 Teknofest Pre-Incubation Competition**: Best Startup Award in Aerospace & Defense
- **2023 Turkish Airlines and Turkish Technic Design Hackathon**: Finalist
- **2023 Teknofest Efficiency Challenge Electric Vehicle Races**: 2nd Place in Hydromobile Category
- **2023 Boeing x Eskişehir Technical University Future of Aviation Competition**: 2nd Place

Skills

Software & Simulation: MATLAB/Simulink, PSIM, Python
Hardware & Design: Altium Designer, LTspice, PSpice, Ansys
Aeronautics & Mechanics: XFLR5, COMSOL, SolidWorks